

Project Title:**Identifying High-Potential but Under-Engaged Customers**

This project aims to identify high-potential customers who are currently under-engaged using customer segmentation techniques. The main research question is whether behavioral and demographic data can be used to detect such customers and improve targeting strategies.

The primary dataset will be the Customer Personality Analysis Dataset, which includes demographic variables (age, income, education) and behavioral variables (purchase frequency, recency, product-level spending, and campaign responses). The dataset is confirmed to represent customers from the United States.

To enrich the dataset, monthly US inflation (CPI) data will be integrated by matching each customer's membership date with the corresponding economic conditions. In addition, an external salary dataset will be used to construct expected income benchmarks based on education level, allowing comparison of each customer's income relative to peers.

The analysis will include data preprocessing, exploratory data analysis (EDA), and feature engineering, including RFM metrics and relative income measures. Customer segmentation will be performed using clustering methods such as K-Means, and clusters will be evaluated using the Elbow Method and Silhouette Score.

The expected outcome is to identify actionable customer segments, particularly high-potential but under-engaged customers, and provide data-driven targeting strategies to improve marketing efficiency.